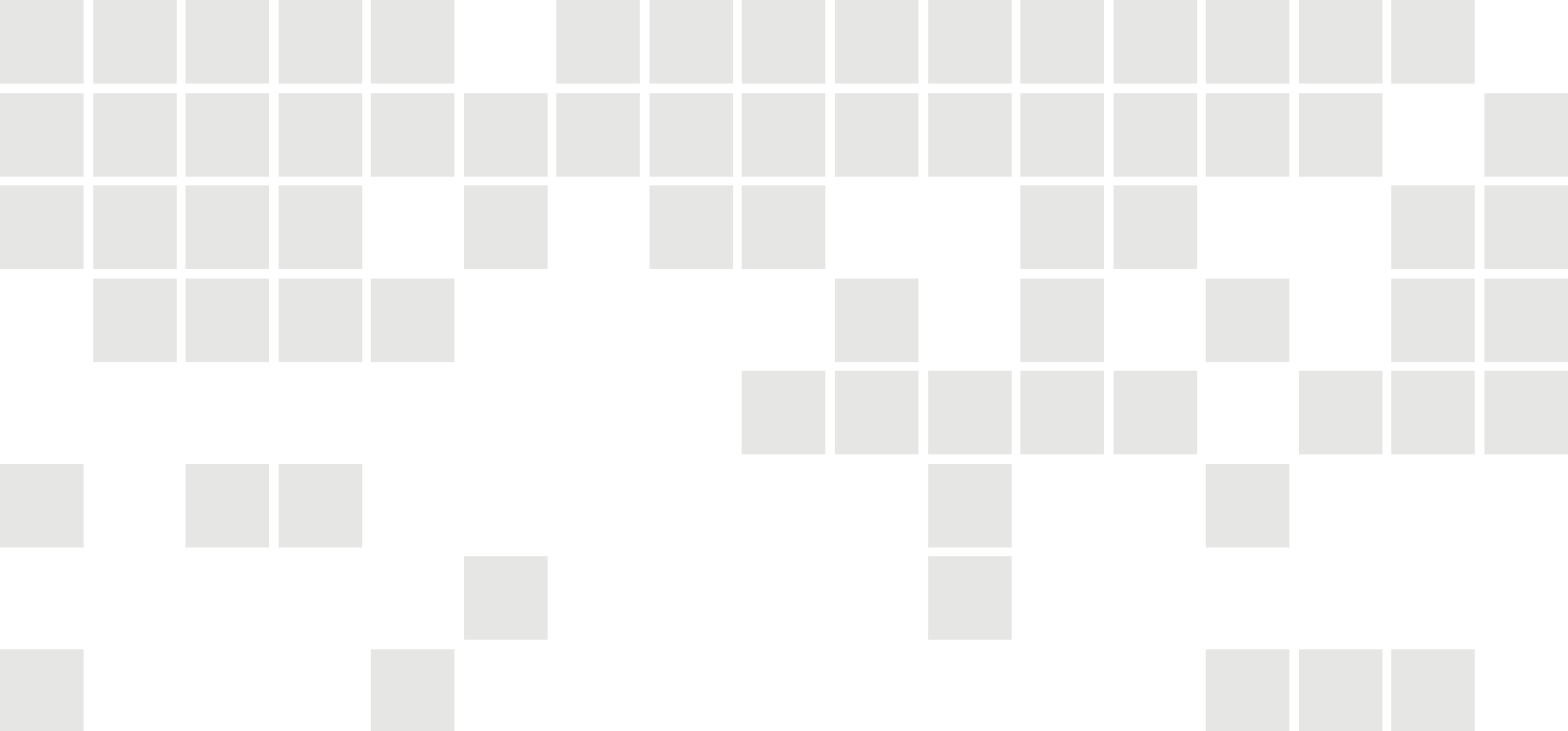




USRSE 2026



Advancing Science in the Age of AI

San Jose, CA · October 19–21, 2026



Full program details and abstracts

<https://us-rse.org/usrse26/program/>

Welcome!

Placeholder. The General Chairs replace this file with their welcome letter before the program is printed, and remove `placeholder: true` above. Plain Markdown paragraphs, *emphasis*, **strong**, and links are supported; links print as their text.

Alex Koufos and Keith Beattie **General Chairs, USRSE'26**

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Schedule Overview

Full abstracts and presenter names are in the online program: <https://us-rse.org/usrse26/program/>

Monday, October 19

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Welcome and Plenary Session , <i>San Jose Ballroom Salon 3 & 4</i> Keynote — Fernando Pérez
10–10:30am	Morning Break , <i>Anywhere</i>
10:30–11:30am	Panel , <i>San Jose Ballroom Salon 3 & 4</i> Student & Early Career — Careers in RSE: Experiences from Research Software Engineers Across Industry, Academia, and Laboratories
10:30am–12pm	RAG Assistants , <i>San Jose Ballroom Salon 1 & 2</i> • AquiLLM: Evaluating Faithfulness in Open-Weight RAG-LLM Systems for Scientific Research • LZ Oracle: an AI-assisted knowledge retrieval platform for the LUX-ZEPLIN dark matter experiment • Data Sherpa: An AI-Powered Assistant for Scientific Collaboration Knowledge Management • AskALCF: A RAG Assistant for HPC User Support Case Studies Demonstrating RSE Impacts Part I , <i>San Jose Ballroom Salon 5 & 6</i> • Automating Data-Driven Alternative Text in Quarto-based NOAA Fisheries Reports • A Modular Threaded Software Application for Axion Detection with DMRadio • Poisonous Books: Rule Based Physics Pipeline and Agentic LLMs for Detecting and Analyzing Toxic Pigments in Victorian Bookbindings • Building Modern Tools for Computational Science in Flutter to Track the American Chestnut AI Assisted Code Development , <i>Willow Glen Room</i> • Reuse, Compose, Extend, Standardize, Automate: Two Decades of RSEing Open (Neuro)Science at CON • Research Software Engineering at SNS and HFIR: Advancing Neutron Science Software in the Age of AI-Assisted Development • Designing for Trust in AI-Assisted Scientific Computing
12–1:30pm	Lunch Break , <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Team Operations , <i>San Jose Ballroom Salon 3 & 4</i> Birds of a Feather — Navigating RSE Team Operations: Leadership, Sustainability, and Emerging Challenges Institutional RSE Community , <i>San Jose Ballroom Salon 1 & 2</i> Birds of a Feather — Building an Institutional RSE Community Better Scientific Software Fellowship , <i>San Jose Ballroom Salon 5 & 6</i> Birds of a Feather — Better Scientific Software Fellowship Community Threat Modeling for Research Software Systems , <i>Willow Glen Room</i> Workshop — A Group Attempt at Threat Modeling for Research Software Systems
3–3:30pm	Afternoon Break , <i>Anywhere</i>

Monday, October 19, continued

3:30–5pm	Interactive Computing and Reproducible Workflows, <i>San Jose Ballroom Salon 1 & 2</i> • Effective Tutorial Notebooks for Cloud Computing Workflows using Remote Sensing Datasets • Marimo: an open-source reactive notebook for Python • Beyond human readers: designing domain-specific language notebooks for AI-assisted code generation • From Notebook to Control Room: Reproducible MLOps for Accelerator Digital Twins Lessons from Training Users & RSEs, <i>San Jose Ballroom Salon 5 & 6</i> • INTERSECT After Three Years: Lessons Learned from Delivering Advanced Research Software Engineering Bootcamps • Beyond Workshops: RSE-Led Hackathons as Experiential Learning for Software Practice in the AI Era • Building Open Source Communities Through Effective Tutorials • Evolving HPC User Support: Just-in-Time Learning with Instructional Videos AI + Software Quality, <i>Willow Glen Room</i> • Software Sustainability Toolkit: Measuring and Improving Sustainable Research Software at Scale • Measuring What Matters in the Age of AI: The Research Software Metrics Landscape • Back to Basics: Design-First Development in the Age of AI Coding Assistants • Scientific Open-Source Software Is Less Likely to Become Abandoned Than One Might Think: Lessons from Curating a Catalog of Maintained Scientific Software
5:30–7:30pm	Poster Reception, <i>San Jose Ballroom Salon 3 & 4</i>

Tuesday, October 20

7:30–8:30am	Registration, <i>Prefunction</i>
8:30–10am	Plenary II, <i>San Jose Ballroom Salon 3 & 4</i>
10–10:30am	Morning Break, <i>Anywhere</i>
10:30–11:30am	Making AI Sustainable: Token Economics, <i>San Jose Ballroom Salon 3 & 4</i> • Agents on a Ramen Budget: Token-Smart Agentic Workflows with Local Models • TBD
10:30am–12pm	RSE Practices in the Age of AI Part I, <i>San Jose Ballroom Salon 1 & 2</i> • Research Software Engineering in the Age of Generative AI • Beyond Editing Code: Uses for AI Tools in Scientific Work • Ten Projects, Two Teams, One Question: How Does AI Impact Development? • Coding is Easy, Everything else is Hard: Lessons learned from working with third-party software developers The Human Factor, <i>San Jose Ballroom Salon 5 & 6</i> • Usability Challenges across AI-Assisted Infrastructure, Security, and Scientific Workflows • From Mood Boards to Web Components: Our Journey Building a Design System for Research • Human in the Loop: Requirements Gathering and Researcher Trust in the Age of AI • Designing for Two Audiences: AI-Readiness and Human Data Usability HPC + AI, <i>Willow Glen Room</i> • DART: Distributed Assignment of Research Tasks for Heterogeneous Compute Environments • Ten Practices for Refactoring Bioinformatics Pipelines to nf-core DSL2 on Shared Slurm HPC • Tapis AI Assistant: Increasing Research Impact through Traceable and Evaluated Agentic Tools • Structuring agentic AI for HPC code modernization

Tuesday, October 20, continued

12–1:30pm	Lunch Break , <i>San Jose Ballroom Salon 3 & 4</i>
12:45–1:15pm	Working Group Fair (Presentations during lunch) , <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Skill Training, <i>San Jose Ballroom Salon 3 & 4</i> Birds of a Feather — How to train an RSE: A discussion of RSE skill training - past, future, and present User Experience: Journey Mapping, <i>San Jose Ballroom Salon 1 & 2</i> Workshop — Journey Mapping: A technique to see opportunities and recognize gaps in user experience Improve Research Outputs by Improving Institutional Collaboration, <i>San Jose Ballroom Salon 5 & 6</i> Birds of a Feather — How can institutional RSE hubs and OSPOs work together for the betterment of research outputs? Refactoring Code for Reuse and Impact, <i>Willow Glen Room</i> Workshop — Fixer-Upper Software: Refactoring Code for Reuse and Impact
3–3:30pm	Afternoon Break , <i>Anywhere</i>
3:30–5pm	Case Studies Demonstrating RSE Impacts Part II, <i>San Jose Ballroom Salon 1 & 2</i> • Josh: Efficient, Portable, and Scalable Cross-Disciplinary Vegetation Modeling via Domain-Specific Language • Take it on the road: How RCD Helps Field Researchers Report Culvert Condition even when Offline • Escaping the Enclave: Format-Faithful Synthetic Data for Open, Reproducible Medicare Claims Pipelines • Automated Data Reduction for Neutron Scattering Experiments: Current Capabilities and Future Directions Code Modernization Using AI, <i>San Jose Ballroom Salon 5 & 6</i> • Agentic code modernization • Scientific Agentic Engineering: A Framework for Reliable AI Agents in Research Software • From Laptop to Cluster: HPC Translation Patterns for AI Stacks • Beyond Code Pairs: Generating Test-Grounded Data for AI-Assisted Scientific Software Modernization Workflows, <i>Willow Glen Room</i> • Cloud-Native Graphical and Programmatic Cross-Database Access to Experiment Metadata with Trino, Superset and Ibis • Custom Containers for Continuous Integration via Spack: Decoupling Environment Configuration from Test Execution • OSC-IS: Research Software Infrastructure for Scientific Artifact Preservation, Provenance, and Reproducible Collaboration • Fabla: An Open-Source Voice-First EMA Platform for Clinical Research

Wednesday, October 21

7:30–8:30am	Registration , <i>Prefunction</i>
8:30–10am	Plenary III & Sponsor Talks, <i>San Jose Ballroom Salon 3 & 4</i> Announcements, US-RSE Updates, US-RSE Awards, BSSw Fellowship Awards
10–10:30am	Morning Break , <i>Anywhere</i>

Wednesday, October 21, continued

10:30am–12pm	Collaboration & Community Building, <i>San Jose Ballroom Salon 1 & 2</i> • Building Inclusive Talent Pathways: Scaling Outreach, Workforce Development, and Community Engagement for RSE • HPC Carpentry and Community Building at NIST • Navigating the Code Review Process as a Team: A Case Study • Collaboration within RSE group at Princeton: development of slurmise for resource usage estimation Software Quality, <i>San Jose Ballroom Salon 5 & 6</i> • Developing a Handbook of best practices for research software sustainability at Princeton • We Create Quality: Towards a Human-Centric Theory of Research Software Quality in the Age of AI • Beyond Productivity: Human-Centered Guardrails for AI-Assisted Research Software Workflows • Performance Portable Software Design Patterns – What they can do and why we should collect them. Generative AI + Applications, <i>Willow Glen Room</i> • From Chaos to Schema: LLM-Powered Document Information Extraction • Steering an LLM AutoResearch Loop with Domain Context: A Case Study with Vehicle Energy Models • Logic Meets Intuition: The Neuro-Symbolic Upgrade for LLMs
12–1:30pm	Lunch Break, <i>San Jose Ballroom Salon 3 & 4</i>
1:30–3pm	RSE Practices in the Age of AI Part II, <i>San Jose Ballroom Salon 1 & 2</i> • AI-Augmented Research Software Engineering: Risks, Challenges, and Practices • A Framework for Evaluating AI-Integrated Research Software: Demonstration and Validation through CyberFacCES Smart Search • A Nonprofit's Google Summer of Code Experience in the Age of AI-Assisted Contributions • Preparing Students for AI-Assisted Research Workflows Reflections on RSE Roles & Pathways, <i>San Jose Ballroom Salon 5 & 6</i> • What They Don't Tell You About the Data Science to RSE Transition • Bears, Bottlenecks, and Bus Factors: Why Embedded RSEs Pay Off • Mission Control: Service and the Stewardship of Science Building Infrastructure for Communities or Organizations, <i>Willow Glen Room</i> • From Lakehouse to Labmate: KBase's Vision for AI-Enabled Scientific Discovery • HEROIC: A Research Software Infrastructure for Real-Time Multi-Messenger Observatory Coordination • A Managed Database-as-a-Service Platform for Scientific Computing • One Size Fits None: Rethinking Campus Research Computing
3–3:15pm	Transition Break, <i>Anywhere</i>
3:15–4:30pm	Rapid Access Microtalks (RAM), Closing Remarks, <i>San Jose Ballroom Salon 3 & 4</i>

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Keith Beattie, Lawrence Berkeley National Laboratory

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Technical Program Committee

Technical Program Chairs

Drew Paine, Lawrence Berkeley National Laboratory

Hilario Torres, NSF NCAR

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Talks: Adam Rubens (NYU Langone Health), Anas AlSobeh

Notebooks: Yuxi (Ethan) Cui, Primus Kabuo

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